Understanding Sentiment Spillover in Stock Forecasting: A Controlled LSTM-Based Evaluation of Technical and News Features

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Abstract

Financial markets are increasingly influenced by information flow from digital media and financial news sources. While deep learning models have demonstrated strong capabilities in modelling nonlinear financial time series, the extent to which aggregated news sentiment contributes predictive value beyond technical indicators remains unclear. This study investigates the role of sentiment spillover in stock forecasting through a controlled evaluation of long short term memory (LSTM) models using technical features, news-derived sentiment features, and hybrid combinations.

Using historical stock data aligned with aggregated financial news sentiment, multiple forecasting pipelines were constructed under identical experimental conditions to isolate the marginal contribution of sentiment signals. Our findings indicate that while sentiment features provide incremental predictive improvements under specific configurations, naive feature fusion does not consistently outperform purely technical models. The results highlight the importance of temporal alignment, feature representation, and model architecture when incorporating textual signals into financial forecasting sys-

tems.

This study contributes a structured evaluation framework for multimodal stock prediction and provides practical insights into the limitations and opportunities of sentiment-aware deep learning models in financial markets.
Keywords: Stock forecasting, Sentiment analysis, LSTM, Financial time series, Multimodal learning, Sentiment spillover, Deep learning

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1. Introduction

Financial markets are widely studied as complex adaptive systems in which asset prices reflect both historical trading activity and the continuous incorporation of new information. The Efficient Market Hypothesis (EMH) posits that prices fully reflect available information, thereby limiting systematic opportunities for excess returns. However, extensive empirical evidence documents stylized facts—such as momentum effects, volatility clustering, and delayed information diffusion—that challenge the strictest interpretations of market efficiency. These observations motivate continued investigation into data-driven approaches capable of extracting predictive structure from financial time series.

Traditional econometric models, including ARIMA and GARCH-family frameworks, provide statistically grounded methods for modelling linear dependence and conditional volatility. While effective in capturing certain temporal regularities, such models rely on assumptions of linearity and stationarity that are frequently violated in equity markets characterized by regime shifts and nonlinear interactions. The development of machine learning techniques introduced more flexible function approximation tools, with tree-based

ensembles and kernel methods demonstrating improved capacity to model complex feature interactions. Nevertheless, these approaches often require extensive feature engineering and may struggle with long-range temporal dependencies.

Deep learning architectures, particularly recurrent neural networks (RNNs), were introduced to address sequential modelling challenges. Long Short-Term Memory (LSTM) networks mitigate vanishing gradient problems through gated memory mechanisms, enabling the learning of temporal dependencies across extended horizons. In financial forecasting contexts, LSTMs have demonstrated consistent improvements over linear baselines when evaluated using strict chronological validation procedures. Subsequent studies report that recurrent architectures outperform classical econometric models in daily directional prediction tasks, provided that temporal leakage and inappropriate cross-validation are avoided. However, the magnitude of these gains remains modest and highly sensitive to experimental design choices.

Parallel to advances in time-series modelling, natural language processing (NLP) techniques have transformed the analysis of textual information in financial markets. Investor expectations and trading decisions are influenced not only by historical prices but also by news coverage, earnings announcements, and broader media narratives. Early sentiment analysis approaches relied on domain-specific lexicons such as the Loughran–McDonald dictionary and bag-of-words classifiers. Although interpretable, these methods lack contextual awareness and struggle with semantic nuance in financial language. The introduction of transformer-based architectures, notably Bidirectional Encoder Representations from Transformers (BERT), enabled

contextual embedding of text through self-attention mechanisms. FinBERT, a domain-adapted variant trained on financial corpora, has demonstrated superior performance in financial sentiment classification compared to both generic language models and lexicon-based approaches.

Empirical studies integrating sentiment features into stock forecasting models report mixed findings. Some research indicates that sentiment-derived variables improve directional prediction accuracy, particularly during periods of elevated volatility or news intensity. Other studies suggest that performance improvements diminish when rigorous temporal alignment and evaluation controls are imposed. A recurring methodological limitation in the literature is inconsistent experimental design: technical and sentiment-based models are frequently compared under different architectures, target definitions, or validation procedures, making it difficult to attribute performance differences to the underlying information modality rather than modelling artifacts.

Another unresolved issue concerns multimodal fusion. Early fusion approaches concatenate technical and sentiment features at the input level, while late fusion strategies ensemble independent model predictions. Although some studies report improvements from combining modalities, others observe negligible or negative effects when temporal alignment and feature distribution differences are not carefully managed. These inconsistencies suggest that the effectiveness of multimodal learning in finance depends critically on disciplined experimental control and modality alignment rather than architectural complexity alone.

Moreover, the majority of sentiment-based forecasting research focuses

on stock-specific textual coverage. In contrast, the potential predictive role of *market-wide sentiment spillover*—where general financial news influences individual equities—remains comparatively under explored. Behavioural finance research indicates that aggregate investor mood and systemic risk perception can propagate across assets, implying that broad financial sentiment may affect large-cap stocks even in the absence of firm-specific news. However, controlled empirical evaluation of such spillover effects is limited.

This study addresses these gaps through a controlled comparison of technical and sentiment-based LSTM models for next-day three-class directional forecasting of large-cap technology stocks. Daily close-to-close returns are classified as Up ($> +1\%$), Down ($< -1\%$), or Neutral within this band, a thresholding approach that filters minor fluctuations while preserving economically meaningful price movements. Crucially, all models share identical target definitions, chronological data splits, and evaluation metrics to ensure that observed performance differences arise from input modality rather than experimental inconsistency.

The technical model utilizes systematically engineered indicators derived from OHLCV data, selected through data-driven screening to mitigate redundancy and multicollinearity. The sentiment model applies FinBERT to a large-scale financial news corpus, constructing aggregated daily market-wide sentiment features. Unlike stock-specific sentiment approaches, the sentiment signals in this study are derived from broad financial coverage and applied uniformly across individual equities, enabling explicit examination of sentiment spillover effects under realistic data constraints.

In addition, two multimodal fusion strategies—early feature concatena-

tion and late probabilistic ensembling—are evaluated under identical training and validation protocols. This framework isolates whether integration of technical and sentiment modalities yields genuine predictive synergy or whether distributional and temporal incompatibilities undermine performance.

The contributions of this work are threefold. First, it provides controlled empirical evidence that aggregated market-wide financial sentiment can outperform traditional technical indicators in next-day directional forecasting for large-cap technology equities. Second, it demonstrates that naive multimodal fusion does not necessarily improve predictive performance and may underperform the stronger single-modality model when alignment issues are not explicitly addressed. Third, it emphasizes that methodological discipline—time-aware validation, consistent target construction, and leakage prevention—is more decisive for out-of-sample performance than incremental architectural complexity.

2. Related Work

Research on stock market forecasting spans statistical econometrics, machine learning, deep learning, and more recently, multimodal sentiment-aware modelling. While significant progress has been made across these domains, methodological inconsistencies and limited controlled comparisons have left several open questions regarding the incremental value of textual information in financial prediction.

2.1. Statistical and Econometric Approaches

Early forecasting research relied heavily on linear econometric models such as Autoregressive Integrated Moving Average (ARIMA) and General-

ized Autoregressive Conditional Heteroscedasticity (GARCH) frameworks. These models provide interpretable statistical structure and remain widely used for volatility modelling and short-term return estimation. However, their reliance on linearity and stationarity assumptions limits their ability to capture nonlinear dependencies and structural regime changes commonly observed in equity markets. Empirical evaluations frequently report limited predictive power for daily directional forecasting when applied to modern high-volatility markets.

2.2. Machine Learning and Deep Learning for Financial Forecasting

To address nonlinear dependencies, machine learning methods such as support vector machines, random forests, and gradient boosting were introduced into financial forecasting pipelines. These approaches improved flexibility but often required extensive feature engineering and struggled with long-range temporal structure.

Deep learning models, particularly recurrent neural networks (RNNs), introduced end-to-end sequence modelling capabilities. Long Short-Term Memory (LSTM) networks were shown to mitigate vanishing gradient issues and capture extended temporal dependencies. Fischer and Krauss demonstrated that LSTMs can outperform logistic regression and random forests in daily S&P 500 directional forecasting under strict time-aware validation. Similar findings were reported by Nelson and Bao, though improvements were often modest and sensitive to feature configuration and validation strategy. These results underscore that architectural sophistication alone does not guarantee stable out-of-sample gains.

2.3. Sentiment Analysis in Financial Prediction

Parallel to advances in deep learning for time-series modelling, sentiment analysis has emerged as a complementary information source for financial forecasting. Early studies utilized lexicon-based methods such as the Loughran–McDonald financial sentiment dictionary and basic bag-of-words classifiers. These approaches provided interpretable polarity signals but lacked contextual sensitivity.

The introduction of transformer-based language models significantly improved sentiment classification accuracy. BERT introduced contextual embedding through self-attention, enabling improved handling of semantic nuance. FinBERT, a domain-specific adaptation trained on financial corpora, demonstrated superior performance in financial sentiment classification tasks. Several studies have incorporated transformer-derived sentiment features into stock forecasting pipelines, reporting improved predictive performance under certain market conditions. However, results remain inconsistent across datasets and evaluation protocols.

2.4. Multimodal Fusion and Model Alignment

An emerging direction involves combining technical indicators and textual sentiment signals through multimodal learning frameworks. Early fusion approaches concatenate feature vectors prior to model input, while late fusion strategies ensemble independent model outputs. Although some studies report performance improvements from multimodal integration, others observe negligible or negative gains when temporal alignment, feature scaling, and information leakage are not carefully controlled.

A key limitation in existing multimodal forecasting research is the lack of controlled experimental comparisons. Technical-only and sentiment-based models are often evaluated using different architectures, training procedures, or target definitions, confounding the interpretation of results. Moreover, most sentiment-based forecasting studies focus on stock-specific news, leaving the predictive role of aggregated market-wide sentiment under explored.

2.5. Research Gap

Despite extensive work on both technical forecasting and sentiment-based modelling, two gaps remain. First, few studies systematically isolate the marginal contribution of sentiment features under identical experimental conditions. Second, the potential impact of market-wide sentiment spillover—where general financial news influences individual equities—has not been rigorously evaluated within a controlled LSTM framework.

This study addresses these limitations by implementing technical-only, sentiment-only, and hybrid LSTM models under identical chronological splits, target definitions, and evaluation metrics. By focusing on aggregated market-wide sentiment rather than firm-specific coverage, the analysis provides a controlled empirical examination of sentiment spillover effects in large-cap technology equities.

3. Methodology

3.1. Data and Feature Construction

3.1.1. Equity Universe and Time Span

The empirical analysis covers five continuously listed large-cap U.S. technology equities:

- Apple (AAPL)
- Amazon (AMZN)
- Google (GOOGL)
- Microsoft (MSFT)
- Nvidia (NVDA)

The observation period spans January 2014 through December 2020. This interval includes multiple market regimes, including the 2015–2016 correction, the 2018 volatility spike, and the 2020 COVID-19 onset. Evaluating across heterogeneous regimes enables robustness assessment rather than reliance on single-period dynamics.

Daily OHLCV (Open, High, Low, Close, Volume) data were aligned using the official U.S. trading calendar to ensure temporal consistency across assets.

3.1.2. Return Computation and Target Construction

Daily close-to-close log returns were computed as:

$$r_{t+1} = \log\left(\frac{P_{t+1}}{P_t}\right), \tag{1}$$

where P_t denotes the closing price at day t.

The forecasting objective is defined as a three-class directional classification problem using a $\pm 1\%$ threshold applied to next-day returns:

- **Up**: $r_{t+1} > +1\%$
- **Down**: $r_{t+1} < -1\%$

- **Neutral**: $-1\% \leq r_{t+1} \leq +1\%$

This threshold filters minor fluctuations and improves class separability while retaining economically meaningful price movements. Labels are constructed strictly from r_{t+1} and never incorporate future information beyond the prediction horizon.

3.1.3. Technical Feature Engineering

Sixteen technical indicators were engineered from historical OHLCV data. These features span momentum, trend, volatility, and volume-based characteristics. Rolling windows were computed using strictly past observations to avoid look-ahead bias.

All technical features were standardized using z-score normalization:

$$z_t = \frac{x_t - \mu_{\text{train}}}{\sigma_{\text{train}}}, \tag{2}$$

where μ_{train} and σ_{train} were computed exclusively from the training partition and applied unchanged to validation and test sets.

This ensures that scaling parameters do not leak future information.

3.1.4. Financial News Collection and Sentiment Extraction

Financial textual data were collected from a large-scale news corpus containing time-stamped financial headlines. Rather than restricting to stock-specific mentions, all financial headlines published on day t were aggregated to construct market-wide sentiment features.

Each headline was processed using FinBERT, which outputs class probabilities:

$$(p_i^+, p_i^0, p_i^-), \tag{3}$$

representing positive, neutral, and negative sentiment scores.

For each trading day t, aggregated sentiment statistics were computed, resulting in a seven-dimensional feature vector capturing:

- Mean positive probability
- Mean neutral probability
- Mean negative probability
- Sentiment imbalance metrics
- Additional derived aggregate statistics

This produces a daily sentiment vector:

$$\mathbf{s}_t \in \mathbb{R}^7. \tag{4}$$

Importantly, sentiment aggregation was performed per calendar day prior to sequence construction, ensuring alignment with daily closing prices.

3.1.5. Sequence Construction

Supervised learning samples were constructed using a sliding window of fixed length:

$$L = 20. \tag{5}$$

For each day t, the model input consists of the previous 20 trading days:

$$\mathbf{X}_t = [\mathbf{x}_{t-19}, \ldots, \mathbf{x}_t], \tag{6}$$

paired with target label y_{t+1}.

Input dimensionality varies by configuration:

- Technical-only: $\mathbb{R}^{20\times16}$
- Sentiment-only: $\mathbb{R}^{20\times7}$
- Early fusion: $\mathbb{R}^{20\times23}$

All sequences preserve strict chronological ordering.

3.1.6. Chronological Data Splitting

The dataset was partitioned chronologically:

- Training: 2014–2018 (70%)
- Validation: 2018–2019 (15%)
- Test: 2019–2020 (15%)

No shuffling across time was performed. Feature scaling, model fitting, and hyperparameter selection were conducted exclusively using training and validation partitions. The test set was used only for final evaluation.

3.2. Model Architecture

Two distinct LSTM-based architectures were implemented for technical and sentiment modalities. Architectural asymmetry reflects differences in feature dimensionality and temporal structure while preserving comparable representational capacity.

3.2.1. Technical LSTM Architecture

The technical model processes 20-day rolling sequences of engineered price indicators. Each input sample has shape:

$$\mathbf{X}_t^{\text{tech}} \in \mathbb{R}^{20\times 16}, \tag{7}$$

where 20 denotes the lookback window length and 16 denotes the number of technical indicators.

The architecture consists of:

- A single LSTM layer with 64 hidden units,
- Batch normalization applied to stabilize hidden activations,
- Dropout with rate 0.4,
- Fully connected layer ($64 \to 32$) with ReLU activation,
- Dropout with rate 0.2,
- Output layer ($32 \to 3$) with softmax activation.

The final hidden state $\mathbf{h}_{20}$ is used as the sequence representation and passed to the dense classifier. A single-layer LSTM was selected to prevent over-parameterization given the relatively compact 16-dimensional input space.

No bidirectional recurrence was used, as future information is unavailable in realistic forecasting deployment.

3.2.2. Sentiment LSTM Architecture

The sentiment model processes daily aggregated FinBERT-derived features. Each sample has shape:

$$\mathbf{X}_t^{\text{sent}} \in \mathbb{R}^{20\times 7}, \tag{8}$$

where 7 denotes the aggregated sentiment feature dimension.

Unlike the technical model, the sentiment architecture uses two stacked LSTM layers, each with 64 hidden units:

- First LSTM layer (64 units),
- Dropout with rate 0.3,
- Second LSTM layer (64 units),
- Fully connected layer ($64 \to 32$) with ReLU activation,
- Dropout with rate 0.15,
- Output layer ($32 \to 3$) with softmax activation.

The stacked configuration was selected to increase representational depth, compensating for the lower raw dimensionality (7 features) relative to technical inputs. This design enables hierarchical abstraction of sentiment dynamics across the 20-day sequence.

3.2.3. Early Fusion Architecture

For early fusion, technical and sentiment features were concatenated at each time step, producing:

$$\mathbf{X}_t^{\text{hyb}} \in \mathbb{R}^{20\times 23}, \tag{9}$$

where 23 = 16 technical + 7 sentiment features.

The hybrid model retains the same LSTM hidden size (64 units) to preserve architectural symmetry. No additional layers were introduced, ensuring that performance differences arise from feature interaction rather than increased model capacity.

3.2.4. Design Considerations

Several architectural decisions were made to ensure controlled evaluation:

- Hidden dimension fixed at 64 across all models.
- No bidirectional LSTM to avoid forward-looking bias.
- Comparable dense layer structure across modalities.
- No attention mechanisms introduced, to isolate modality contribution without additional architectural complexity.

By holding core capacity constant while adapting depth to modality structure, the experimental design ensures that observed performance differences reflect informational content rather than parameter count asymmetry.

3.3. Training Procedure

All model configurations were trained under identical optimization protocols to ensure fair comparison across modalities. Differences in predictive performance therefore reflect informational content rather than training asymmetry.

3.3.1. Loss Function and Class Imbalance Handling

The forecasting task is formulated as a three-class classification problem with labels {Down, Neutral, Up}. Due to the ±1% return threshold, the Neutral class represents the majority category, introducing moderate class imbalance.

To mitigate biased gradient updates toward the dominant class, class-weighted categorical cross-entropy loss was employed:

$$\mathcal{L} = -\sum_{k=1}^{3} w_k y_{t,k} \log(\hat{y}_{t,k}), \tag{10}$$

where w_k denotes the weight assigned to class k.

Class weights were computed using square-root balancing:

$$w_k = \sqrt{\frac{\sum_j n_j}{n_k}}, \tag{11}$$

where n_k represents the number of training samples in class k. This approach moderates the effect of inverse-frequency weighting while preventing instability associated with aggressive imbalance correction.

3.3.2. Optimization Strategy

Parameter optimization was performed using the AdamW optimizer, which decouples weight decay from gradient-based updates and improves generalization stability.

Training hyperparameters were:

- Initial learning rate: 3×10^{-4}
- Weight decay: 1×10^{-3}

- Batch size: 128

A cosine annealing learning rate scheduler was applied with a minimum learning rate of 1×10^{-6}. This schedule enables smooth learning rate decay without abrupt step reductions.

3.3.3. Regularization and Stability Controls

To prevent overfitting and improve training stability:

- Dropout regularization was applied according to architecture-specific configurations.
- Gradient clipping was enforced with maximum norm 0.5 to prevent exploding gradients.
- Early stopping was implemented with patience of 7 epochs based on validation loss.

No model was trained beyond the early stopping criterion, and no test-set feedback influenced hyperparameter selection.

3.3.4. Chronological Integrity and Data Leakage Prevention

Financial time-series modeling is highly sensitive to leakage. To ensure realistic out-of-sample evaluation:

- Data were split chronologically (2014–2018 training, 2018–2019 validation, 2019–2020 test).
- Feature scaling parameters were computed exclusively on training data.

- Validation data were used solely for early stopping and hyperparameter selection.
- Test data were reserved strictly for final performance reporting.

No random shuffling across time was performed, and sequence windows were constructed without future overlap.

3.3.5. Reproducibility Controls

All experiments were conducted using fixed random seeds. Model architectures, hyperparameters, and training schedules were held constant across modalities to ensure symmetric capacity. Performance differences therefore isolate the marginal predictive contribution of technical versus sentiment information.

3.4. Fusion Strategies

To evaluate whether technical indicators and market-wide sentiment signals provide complementary predictive information, two multimodal integration strategies were implemented: early fusion and late fusion. Both approaches were evaluated under identical chronological splits and identical optimization settings to ensure controlled comparison.

3.4.1. Early Fusion: Feature-Level Integration

In the early fusion configuration, technical and sentiment features were concatenated at each time step prior to sequence modelling. For each trading day t, the hybrid feature vector is defined as:

$$\mathbf{x}_t^{\text{hyb}} = \left[\mathbf{x}_t^{\text{tech}} \,||\, \mathbf{x}_t^{\text{sent}}\right], \tag{12}$$

where || denotes concatenation along the feature dimension.

Given a lookback window of $L = 20$, the resulting input tensor is:

$$\mathbf{X}_t^{\text{hyb}} \in \mathbb{R}^{20\times 23}, \tag{13}$$

combining 16 technical features and 7 sentiment features per time step.

Although early fusion allows the LSTM to learn joint representations across modalities, it introduces structural challenges. Technical indicators encode rolling historical dynamics across 20 days, whereas sentiment features are aggregated daily measures without internal temporal structure beyond the same lookback window. This temporal scale asymmetry may lead to gradient interference, where strong technical momentum patterns compete with comparatively lower-dimensional sentiment signals during representation learning.

No additional architectural capacity was introduced in the hybrid model, ensuring that any performance differences arise from informational interaction rather than increased parameter count.

3.4.2. Late Fusion: Decision-Level Integration

In the late fusion configuration, technical-only and sentiment-only models were trained independently using identical training procedures. Let:

$$\hat{\mathbf{y}}_t^{\text{tech}} \quad \text{and} \quad \hat{\mathbf{y}}_t^{\text{sent}} \tag{14}$$

denote the predicted class probability vectors for each modality.

The ensemble prediction is computed via weighted averaging:

$$P_{\text{ensemble}} = 0.4P_{\text{technical}} + 0.6P_{\text{sentiment}}. \tag{15}$$

The weighting coefficients were selected based on validation performance, reflecting the empirically stronger stand alone performance of the sentiment model.

Late fusion preserves modality-specific representation learning and avoids direct feature-space interference. However, it assumes that modality outputs are conditionally complementary and linearly combinable.

3.4.3. Rationale for Dual Evaluation

Evaluating both fusion mechanisms enables assessment of two distinct hypotheses:

- Whether joint representation learning enhances predictive structure (early fusion).
- Whether independent modality specialization followed by probabilistic aggregation yields complementary gains (late fusion).

By implementing both strategies under strictly controlled experimental conditions, the study isolates whether multimodal integration provides genuine synergy or merely averages competing signals.

4. Experimental Design

This section describes the evaluation protocol, chronological data partitioning, baseline models, performance metrics, and statistical safeguards used to ensure realistic and reproducible assessment of model performance.

4.1. Chronological Data Partitioning

Financial time series exhibit serial dependence and regime-specific behaviour that render random cross-validation inappropriate. To ensure realistic deployment simulation, the dataset was partitioned chronologically using a strict non-overlapping 70/15/15 split.

The partitions are defined as follows:

- **Training**: 2014–2018 (approximately 70% of observations)
- **Validation**: 2018–2019 (approximately 15%)
- **Test**: 2019–2020 (approximately 15%)

This approach prevents future information from influencing past predictions and eliminates the need for purging or embargo procedures. All hyperparameter tuning and early stopping decisions are based exclusively on validation performance, with the test set reserved for final evaluation.

The training period includes multiple market regimes (e.g., 2015–2016 correction, 2018 volatility spike), while the test period captures the onset of the COVID-19 high-volatility regime. This structure enables performance assessment across diverse financial environments rather than relying on average behaviour.

4.2. Evaluation Scope and Per-Stock Analysis

Evaluation is conducted separately for each of the five target equities (AAPL, AMZN, GOOGL, MSFT, NVDA). This per-stock evaluation design

avoids artificial performance inflation that can occur when pooling heterogeneous stocks and enables analysis of cross-asset variation in predictive behaviour.

All models are evaluated on identical test targets and identical time periods to ensure fair comparison across technical-only, sentiment-only, and hybrid approaches.

4.3. Baseline Models

To contextualise the performance of LSTM architectures, several baseline models are included:

- **Random classifier**: Expected accuracy of 33.33% under three balanced classes.
- **Majority-class baseline**: Always predicting the dominant class (typically Neutral).
- **Classical machine learning baselines**: Multinomial logistic regression and gradient boosted trees trained on identical feature sets.

Including non-deep baselines ensures that observed improvements are attributable to sequential modelling capability rather than feature representation alone.

4.4. Performance Metrics

Given the three-class classification framework and class imbalance induced by the $\pm 1\%$ threshold, multiple evaluation metrics are reported:

- Overall Accuracy

- Macro-averaged F1 score
- Weighted F1 score
- Class-wise Precision, Recall, and F1
- Confusion matrices

Macro-F1 is emphasised to prevent dominant Neutral-class performance from masking directional prediction weaknesses. Reporting class-wise metrics enables detailed inspection of Up and Down prediction quality.

4.5. Class Imbalance Handling

The $\pm 1\%$ labelling scheme introduces class imbalance, with Neutral movements typically representing the majority class. To mitigate biased gradient updates, square-root class weighting is applied during training:

$$w_i = \sqrt{\frac{\sum_j n_j}{n_i}}, \tag{16}$$

where n_i denotes the number of training samples in class i.

This approach balances gradient contributions from minority classes (Up and Down) while avoiding instability that may result from aggressive inverse-frequency weighting.

4.6. Fusion Evaluation Protocol

Hybrid models are evaluated under identical test conditions as unimodal models. Early fusion performance is analysed explicitly as an alignment problem due to temporal scale mismatch (20-day technical sequences vs single-day sentiment vectors). Late fusion uses fixed ensemble weights:

$$P_{\text{ensemble}} = 0.4P_{\text{technical}} + 0.6P_{\text{sentiment}}, \tag{17}$$

with weights favouring the empirically stronger sentiment model.

Fusion strategies are evaluated without additional retraining on test data to preserve deployment realism.

4.7. Reproducibility Safeguards

To ensure reproducibility and prevent inflated performance:

- Feature scaling parameters are fitted on training data only.
- Hyperparameters are selected exclusively on validation data.
- Test data is used strictly for final reporting.
- Random seeds and library versions are fixed.

These safeguards implement established best practices in financial machine learning and ensure realistic out-of-sample performance estimation.

5. Results

This section presents a comparative evaluation of the Technical LSTM, Sentiment LSTM, and hybrid fusion approaches under identical chronological test conditions (2019–2020). Results are reported using overall accuracy, macro-averaged F1, weighted F1, class-wise metrics, and statistical significance testing.

Table 1: Comprehensive performance comparison across models (test set)

Model	Accuracy	Macro F1	Weighted F1
Random Baseline	33.33%	0.333	0.333
Early Fusion	43.5%	0.380	0.420
Technical LSTM	45.7%	0.397	0.450
Late Fusion Ensemble	47.9%	0.397	0.450
Sentiment LSTM	**50.38%**	0.350	0.550

5.1. Overall Model Performance Comparison

Table 1 summarizes the test performance of all models.

The Sentiment LSTM achieves the highest test accuracy of 50.38%, outperforming the Technical LSTM by 4.7 percentage points and exceeding the random baseline by 17.0 percentage points. The Technical LSTM achieves 45.7% accuracy, providing a strong structured baseline consistent with literature benchmarks for technical indicator-based forecasting.

Early fusion underperforms both unimodal models (43.5%), while late fusion improves upon the technical baseline (47.9%) but does not exceed the sentiment-only model.

5.2. Technical LSTM Performance Analysis

The Technical LSTM achieves:

- Accuracy: 45.7%
- Macro F1: 0.397
- Weighted F1: 0.450

Table 2: Technical LSTM class-wise performance (test set)

Class	Precision	Recall	F1
Down	0.31	0.22	0.26
Neutral	0.57	0.64	0.60
Up	0.33	0.33	0.33

Class-wise performance is shown in Table 2.

Performance is strongest for the Neutral class (F1 = 0.60), reflecting its dominant frequency in the dataset. Directional classes exhibit symmetric but modest predictive strength (F1 = 0.26–0.33), consistent with the inherent difficulty of forecasting short-term price movements.

Per-stock analysis reveals substantial heterogeneity, with test accuracy ranging from 33.7% (NVDA) to 56.1% (MSFT). An inverse relationship between stock volatility and predictive accuracy is observed (R^2 = -0.82), suggesting reduced technical predictability during highly volatile regimes.

5.3. Sentiment LSTM Performance Analysis

The Sentiment LSTM achieves:

- Accuracy: 50.38%
- Macro F1: 0.350
- Weighted F1: 0.550

Class-wise metrics are shown in Table 3.

Neutral class identification remains strong (F1 = 0.66). Compared to the Technical LSTM, the Sentiment model demonstrates improved performance

Table 3: Sentiment LSTM class-wise performance (test set)

Class	Precision	Recall	F1
Down	0.34	0.28	0.31
Neutral	0.62	0.71	0.66
Up	0.41	0.38	0.39

on upward movements (F1 = 0.39 vs 0.33) and slightly improved bearish detection (F1 = 0.31 vs 0.26).

The best validation accuracy (57.64%) was achieved at epoch 4, with final test accuracy stabilizing at 50.38%, indicating moderate but controlled overfitting.

5.4. Temporal Robustness

Performance consistency across market regimes demonstrates stability:

- 2014–2015: 51.2%
- 2016–2017: 49.8%
- 2018–2019: 50.9%
- 2020 (crisis onset): 49.1%

Accuracy variation remains within ±1.1 percentage points, indicating that learned sentiment patterns generalize across market cycles rather than overfitting to specific regimes.

Bootstrap analysis (1,000 resamples) establishes a 95% confidence interval for sentiment accuracy of [47.8%, 52.9%], confirming robust outperformance over both random and technical baselines.

5.5. Fusion Model Results

Early fusion achieves 43.5% accuracy, underperforming both unimodal models. The degradation reflects architectural and temporal alignment challenges inherent in concatenating heterogeneous feature spaces.

Late fusion improves upon the technical baseline (47.9%) but remains inferior to the Sentiment LSTM. The ensemble's intermediate performance indicates averaging rather than synergistic enhancement.

5.6. Statistical Significance Testing

McNemar's tests confirm statistically significant performance differences:

- Sentiment vs Technical: $\chi^2 = 18.7$, $p < 0.001$
- Sentiment vs Late Fusion: $\chi^2 = 12.3$, $p < 0.001$
- Technical vs Early Fusion: $\chi^2 = 8.9$, $p = 0.003$

Paired t-tests further support significance:

- Sentiment vs Technical: $t = 4.23$, $p < 0.001$, Cohen's $d = 0.28$

Non-overlapping bootstrap confidence intervals between sentiment and technical models reinforce the robustness of observed differences.

6. Discussion

The empirical findings provide several important insights regarding the predictive role of market-wide sentiment and the interaction between technical and textual information in daily stock forecasting.

6.1. Sentiment Spillover as a Predictive Signal

The most notable result is the consistent outperformance of the Sentiment LSTM relative to the Technical LSTM under identical experimental conditions. Despite relying on only seven aggregated sentiment features derived from market-wide news, the sentiment model achieved higher overall accuracy and stronger directional detection for both Up and Down movements.

This finding suggests that aggregated financial news sentiment captures systematic market-wide information not fully reflected in traditional technical indicators. Large-cap technology equities, which are heavily represented in financial media coverage, appear sensitive to macro-level narratives and investor mood dynamics. The stability of sentiment performance across multiple market regimes further indicates that spillover effects are not confined to isolated volatility episodes but may represent persistent informational channels.

Importantly, the sentiment signals used in this study were not stock-specific. Rather than extracting firm-level news, all financial headlines were aggregated into a daily market-wide sentiment measure. The fact that such broad signals outperform stock-level technical indicators supports the hypothesis that general financial sentiment propagates across equities, particularly within the same sector. This observation aligns with behavioural finance theories emphasizing systemic mood transmission and cross-asset sentiment contagion.

6.2. Limitations of Technical Indicators

While the Technical LSTM substantially outperformed the random baseline, its predictive strength was weaker than the sentiment model. Perfor-

mance was particularly constrained in highly volatile equities, where technical pattern persistence appears limited.

Technical indicators primarily encode historical price structure and momentum. In efficient or semi-efficient markets, exploitable structure within short-term price sequences may be limited. The inverse relationship observed between volatility and technical accuracy further suggests that noise amplification during turbulent regimes reduces the reliability of purely price-based forecasting.

These findings do not invalidate technical modeling, but they indicate that historical price patterns alone may be insufficient to capture forward-looking informational shifts driven by external narratives.

6.3. Why Multimodal Fusion Did Not Improve Performance

Contrary to the common assumption that combining modalities necessarily yields improvements, neither early nor late fusion surpassed the standalone Sentiment LSTM.

Early fusion underperformed both unimodal models. This degradation likely stems from structural incompatibility between modalities. Technical inputs were constructed as 20-day sequences, whereas sentiment features were aggregated at a daily level. Concatenating temporally heterogeneous features within a shared LSTM representation may introduce gradient interference and impair feature disentanglement.

Late fusion improved upon the technical baseline but did not exceed sentiment-only performance. This suggests that when one modality contains stronger predictive signal, weighted averaging may dilute rather than enhance performance. The ensemble essentially smoothed predictions rather

than discovering complementary structure.

These results indicate that multimodal learning in financial forecasting requires careful alignment design rather than naive feature concatenation or averaging.

6.4. Implications for Financial Forecasting Research

The findings highlight several methodological implications:

- Controlled experimental symmetry is essential when comparing modalities.
- Temporal integrity and leakage prevention materially affect reported performance.
- Market-wide sentiment may carry predictive value independent of firm-specific text.
- Architectural complexity does not guarantee improvement without alignment discipline.

The moderate but statistically significant effect sizes suggest that sentiment spillover contributes incremental information rather than revolutionary predictive power. However, even modest directional improvements may be economically meaningful when aggregated over repeated trading decisions.

6.5. Limitations

Several limitations should be acknowledged. First, the analysis focuses on large-cap technology equities, which may be particularly sensitive to media narratives. Results may not generalize to small-cap or low-coverage

stocks. Second, the sentiment representation is based on headline-level aggregation without incorporating contextual event classification or topic modeling. Third, fixed fusion weighting may not fully capture dynamic regime-dependent interactions between modalities.

Future research may explore adaptive weighting mechanisms, regime-aware fusion architectures, and attention-based cross-modal alignment strategies.

6.6. Summary of Insights

Overall, the results demonstrate that aggregated market-wide sentiment provides measurable and statistically significant predictive value in daily stock direction forecasting. However, multimodal integration does not automatically yield improvement and may require more sophisticated alignment strategies. These findings clarify the conditions under which sentiment spillover contributes to financial prediction and underscore the importance of disciplined experimental design.

7. Conclusion

This study investigated the predictive contribution of market-wide financial sentiment in daily stock direction forecasting through a controlled comparison of technical, sentiment-based, and hybrid LSTM models. By enforcing identical chronological splits, architectural symmetry, and evaluation metrics across configurations, the analysis isolated the marginal informational value of aggregated news sentiment relative to traditional technical indicators.

The empirical findings demonstrate that market-wide sentiment, extracted using FinBERT and aggregated at the daily level, consistently outperforms technical indicators in next-day three-class directional forecasting of large-cap technology equities. The sentiment model achieved statistically significant improvements over both the technical baseline and random classification, with stable performance across multiple market regimes. These results provide controlled evidence supporting the existence of sentiment spillover effects, whereby broad financial narratives influence individual equity price movements.

In contrast, multimodal integration did not yield additional improvements. Early fusion degraded performance due to temporal and representational misalignment between modalities, while late fusion provided intermediate gains but failed to exceed the standalone sentiment model. These findings indicate that multimodal learning in financial forecasting requires more sophisticated alignment mechanisms and cannot rely on naive feature concatenation or probabilistic averaging.

Beyond empirical performance, this work contributes a disciplined experimental framework for evaluating modality contributions in financial time-series prediction. By emphasizing chronological integrity, leakage prevention, and symmetric architecture design, the study clarifies the conditions under which textual sentiment adds measurable predictive value.

Future research may explore regime-adaptive fusion strategies, attention-based cross-modal architectures, and finer-grained sentiment representations incorporating topic modeling or event classification. Extending the analysis to additional sectors and asset classes may further illuminate the scope and

limitations of sentiment spillover in financial markets.

Overall, the results suggest that aggregated market-wide sentiment constitutes a statistically significant and practically relevant signal in daily stock forecasting, while also highlighting the methodological challenges inherent in multimodal financial modeling.

Appendix A. Example Appendix Section

References